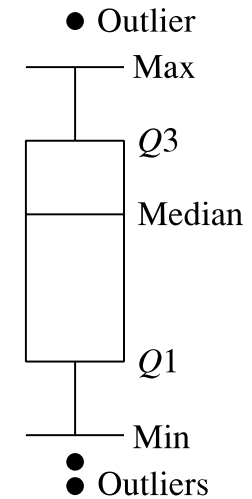


Boxplot Visualization

- ❑ A five-number summary
 - ❑ The smallest nonoutlier value (Min)
 - ❑ The lower quartile (Q1)
 - ❑ The median (Q2)
 - ❑ The upper quartile (Q3), and
 - ❑ The largest nonoutlier value (Max)
- ❑ The interquantile range (IQR) is defined as $Q3 - Q1$
- ❑ Any object that is $1.5 \times \text{IQR}$ smaller than Q1 or $1.5 \times \text{IQR}$ larger than Q3 is treated as an outlier because the region between $Q1 - 1.5 \times \text{IQR}$ and $Q3 + 1.5 \times \text{IQR}$ contains 99.3% of the objects



Grubb's Test

- For each object x in a data set, the z-score is $z = \frac{|x - \mu|}{\sigma}$
 - μ is the mean, and σ is the standard deviation of the input data

- An object x is an outlier if $z \geq \frac{n-1}{\sqrt{n}} \sqrt{\frac{t_{\frac{\alpha}{2n}, n-2}^2}{n-2 + t_{\frac{\alpha}{2n}, n-2}^2}}$

- $t_{\frac{\alpha}{2n}, n-2}^2$ is the value taken by a t-distribution at a significance level of $\frac{\alpha}{2n}$

Detection of Multivariate Outliers

- ❑ The central idea is to transform the multivariate outlier detection task into a univariate outlier detection problem
- ❑ Multivariate outlier detection using the d
 - ❑ For a multivariate data set, let $\bar{\mathbf{o}}$ be the sample mean vector
 - ❑ For an object, $\mathbf{o}$, in the data set, the squared Mahalanobis distance from $\mathbf{o}$ to $\bar{\mathbf{o}}$ is $MDist(\mathbf{o}, \bar{\mathbf{o}}) = (\mathbf{o} - \bar{\mathbf{o}})^T S^{-1} (\mathbf{o} - \bar{\mathbf{o}})$, where S is the sample covariance matrix
 - ❑ Apply Grubb's test to $MDist(\mathbf{o}, \bar{\mathbf{o}})$

Multivariate Outlier Detection Using the χ^2 -statistic

- The χ^2 -statistic is

$$\chi^2 = \sum_{i=1}^n \frac{(o_i - E_i)^2}{E_i}$$

- o_i is the value of o on the i -th dimension
- E_i is the mean of the i -dimension among all objects
- If the χ^2 -statistic is large, the object is an outlier